

Defects: The arrangement of atoms or ions in materials contains imperfections which is termed as defects.

There are 3 types of defects:-

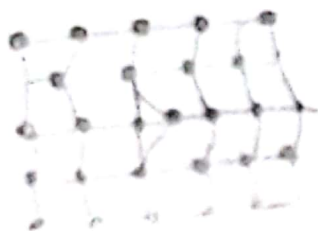
(i) point defects:- Point defects are lattice defects of zero dimensionality, that they don't possess lattice structure at any dimension.

There are two ^{major} types of point defects.

(i) vacancy:- If an atom or ion is missing from its lattice site, the defect caused in the crystal is called a vacancy defect.



(ii) Interstitial defect:- An interstitial defect is formed when an extra atom or ion is inserted into the crystal structure at a normally unoccupied position.



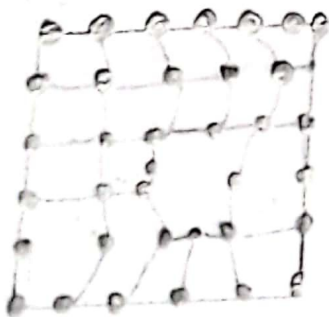
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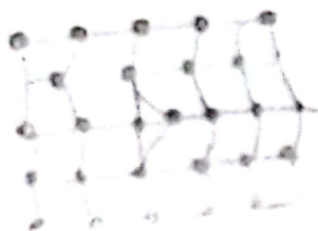
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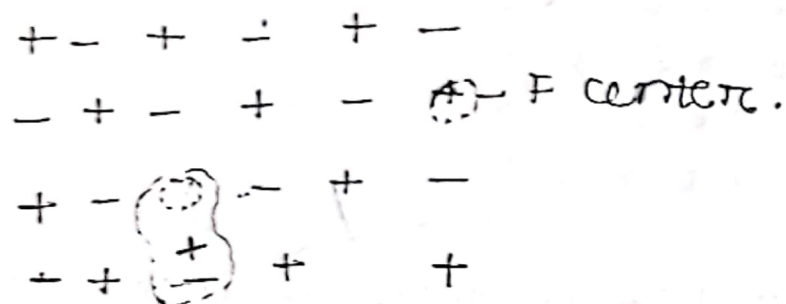
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▣ **Substitutional defect**:- A substitutional defect is introduced when one atom or ion is replaced by a different type of atom or ion.



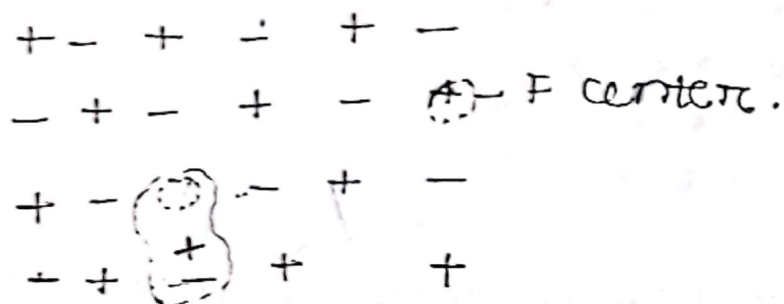
▣ **Frenkel defect**:- A Frenkel defect is a vacancy-interstitial pair formed when an ion jumps from a normal lattice point to an interstitial site, leaving behind a vacancy.



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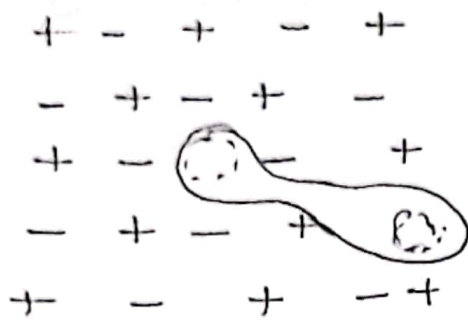


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Schottky defect:- This defect arises when some cations and equal number of anions are missing from their own position and thus leave lattice vacancies or holes in a crystal lattice.



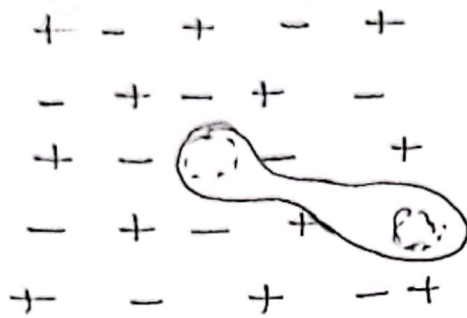
Rules to occur point defect:-

- ①. A charge balance must be maintained so that the crystalline material as a whole is electrically neutral.
- ②. A mass balance must be maintained.
- ③. The number of crystallographic sites must be conserved.

Line defects:- ~~led~~ Lattice distortion which is centered around a line is termed as line defects.

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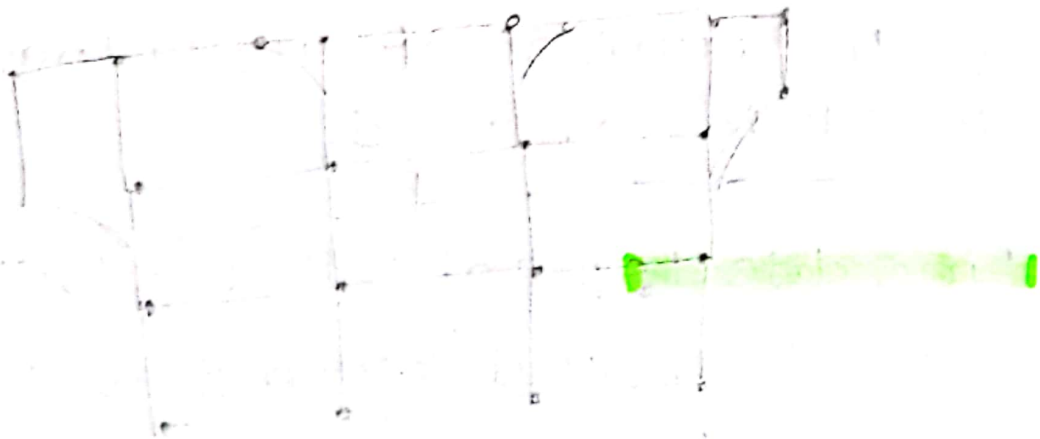
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* Line defects are dislocation.

There are three kinds of dislocation.

(i). Screw; - Screw dislocation is termed as part of set of lattice planes has been moved one or more lattice units relative to their neighbouring lattice plane.

(ii) Edge dislocation; - Edge dislocation is termed as the extra portion of ~~atoms~~ a plane of atoms, or half plane the edge of which terminates the crystal. ~~is the~~

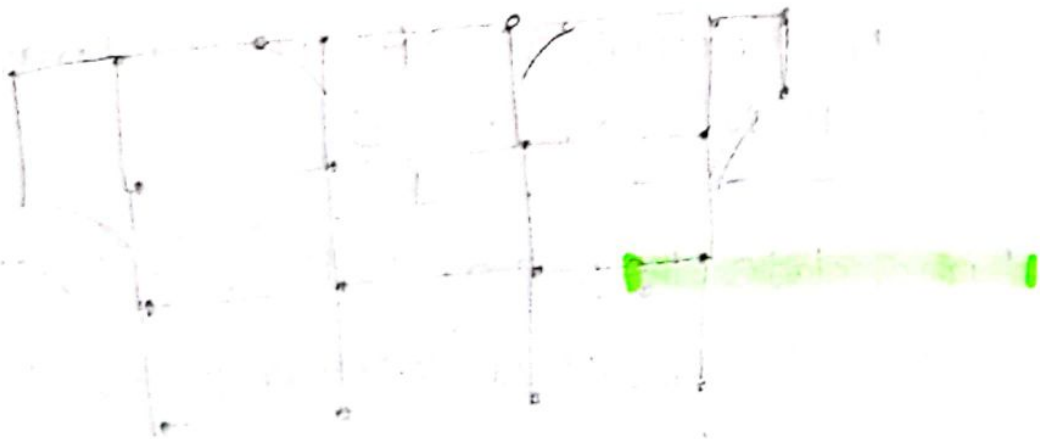


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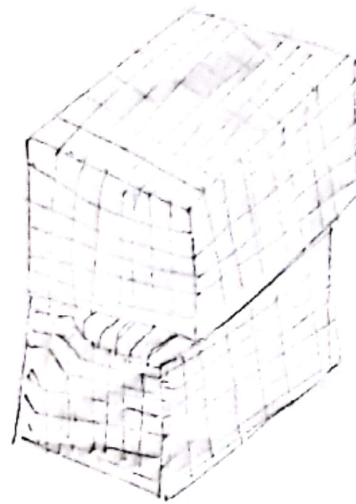
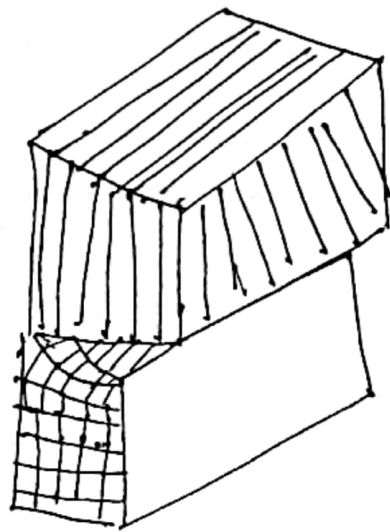
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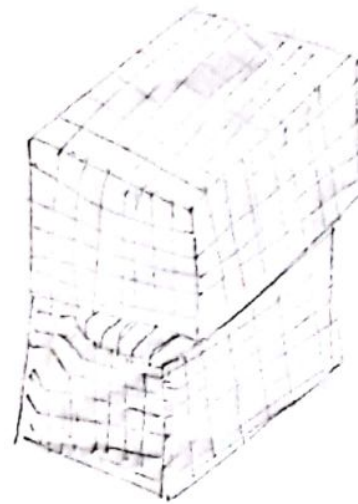
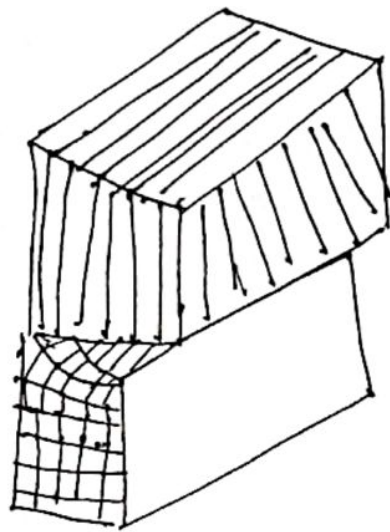
* Description of screw dislocation:-

A screw dislocation can be formed in a perfect crystal lattice. The screw dislocation can be formed in a perfect crystal by applying upward and downward shear stress to region of a perfect crystal that have been separated by a cutting plane. These shear stresses introduce a region of distorted crystal lattice in the form of a spiral ramp of distorted atoms. The region of shear ~~strain~~ strain is created around the screw dislocation in which energy is ~~sto~~ stored. The slip or burger vector is parallel to the dislocation line.



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2021:-

(a) How does defect arise in a crystal?

The defect may arise due to the heat absorbed by the crystal from the surrounding or due to the presence of impurities in the crystal and also due to the irregularities in the arrangement of particle when the rate of crystallization is fast, more defects are present when the rate of crystallisation is slowest, the defects are minimized. The physical and chemical properties crystalline solid are affected due to presence of crystal defects. Defects arise during solidification at high temperatures, or as a consequence of radiation damage. etc

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10 Explain effects of defects on crystalline properties?

- ① Conductivity of extrinsic semi-conductor may be dictated entirely by the trace amount of impurity.
- ② Mechanical and plastic properties of solid are usually controlled by dislocation, a special type of imperfection.
- ③ Diffusion through solid may be accelerated enormously by the presence of impurities.
- ④ Imperfection shows marked influence on thermal and conduction and on hysteresis.
- ⑤ Imperfection in some cases ^{to} produce substitute properties like increase in hardness and strength.

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